

MAHALAXMI SOMISETTY

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OBJECTIVE

Final-year B.Tech Information Technology student with strong foundations in software engineering, data structures, and database systems. Passionate about Generative AI, AI Agents, and building intelligent systems that solve real-world problems. Experienced in developing full-stack and AI-driven applications through hands-on projects and continuous learning.

EDUCATION

B.Tech – Information Technology, VNR VJIET, Hyderabad 2023 – 2027

CGPA: 9.23 / 10.0

Higher Secondary, Sri Chaitanya Junior College, Hyderabad 2021 – 2023

Percentage: 98%

TECHNICAL SKILLS

Languages	C++, Python, JavaScript, SQL, Java
Frontend	React.js, HTML, CSS, Tailwind CSS, Vite, Recharts
Backend	FastAPI, REST APIs, Async Programming, Dependency Injection, JWT
Databases	SQL Server, SQLite, Firebase, ChromaDB, Vector Databases
Data & ETL	ETL Pipelines, Data Warehousing, Medallion Architecture, T-SQL, Star Schema, Data Modeling
AI / Gen AI	RAG, Agentic AI Systems, LangChain, HuggingFace, LLM Integration, Embedding Pipelines, Prompt Engineering, Groq
Tools & Infra	Git, GitHub, Docker (basics), Vercel, Render, VS Code, SSMS, Linux
Concepts	Data Structures & Algorithms, OOP, DBMS, Operating Systems, System Design, REST, HLD

PROJECTS

Autonomous Data Analyst | React.js, FastAPI, LangChain, Groq, ChromaDB, HuggingFace |

[GitHub](#) | [Demo Video](#)

- Built an end-to-end autonomous data analyst powered by a **6-agent LangChain pipeline** (Query Planner → Code Generator → Executor → Error Fixer → Visualizer → Insight Narrator) — users upload any CSV and ask questions in plain English; the system writes Pandas code, self-heals on errors, renders Recharts visualizations, and narrates insights automatically.
- Integrated **Groq (llama-3.3-70b)** as the LLM backbone (free, no credit card) with HuggingFace all-MiniLM-L6-v2 embeddings and ChromaDB as vector store for schema-aware column selection.
- Engineered a **self-healing Executor + Error Fixer agent loop** that retries failed code up to 2 times autonomously, reducing manual intervention and demonstrating real agentic behavior.

RAG + PDF Conversational AI Chatbot | React.js, FastAPI, Vector DB, LLM | [GitHub](#) |

[Live Demo](#)

- Built a production-grade full-stack AI chatbot using **Retrieval-Augmented Generation (RAG)**

- that ingests PDF documents, indexes content into a vector store, and delivers grounded, cited responses — scalable to enterprise document intelligence use cases.

- Implemented **persistent multi-turn conversation history**, enabling the model to maintain context across sessions, replicating the behavior of a natural document Q&A assistant.
- Engineered a streaming React frontend with a FastAPI backend orchestrating document upload, chunking, embedding generation, and vector retrieval pipelines end-to-end.

Latency Lens – API Performance Analyzer | React.js, FastAPI, SQLite | Nov 2025 – Present | [GitHub](#) | [Live Demo](#)

- Developed a full-stack REST API testing and analytics platform supporting GET/POST/PUT/PATCH/DELETE with a real-time dashboard and persistent request history.
- Achieved a **10x reduction in 10-request latency (from ~10s to ~1s)** through concurrent execution; implemented P95/P99 tail latency tracking and SLA threshold alerting for performance outlier detection.
- Architected FastAPI backend with SQLite persistence and deployed React on Vercel and Python backend on Render, adhering to REST API design best practices.

SQL Data Warehouse & Analytics Pipeline | SQL Server, T-SQL, ETL, Medallion Architecture | Oct 2025 | [GitHub](#)

- Architected a **Bronze → Silver → Gold medallion data warehouse** with end-to-end ETL pipelines to ingest, clean, and transform multi-source data into an optimized star schema for analytical querying.
- Authored 20+ complex stored procedures and SQL analytics queries covering KPIs, customer segmentation, year-over-year trends, and executive reporting — designed for large-scale data infrastructure.

COURSES & CERTIFICATIONS

- Generative AI with LangChain & HuggingFace — Udemy (Krish Naik) | RAG pipelines, LLM integration, vector stores, deploying production Gen AI applications
- FastAPI — Udemy | Production-grade REST APIs, async programming, dependency injection, and database integration

ACHIEVEMENTS

- Ranked among the **Top 75 coders in Amazon HackOn** — competed against thousands of students nationwide.
- Maintained a **9.23 CGPA** throughout B.Tech (IT) while independently delivering 4+ end-to-end projects, including a production-deployed full-stack app, a Gen AI RAG chatbot, and an autonomous 6-agent data analyst.
- Solved **250+ coding problems on LeetCode** spanning arrays, binary search, linked lists, stacks, trees, graphs, and dynamic programming.